

AuthEv-LKolb at CheckThat! 2024 Task 5



A Two-Stage Approach To Evidence-Based Social Media Claim Verification

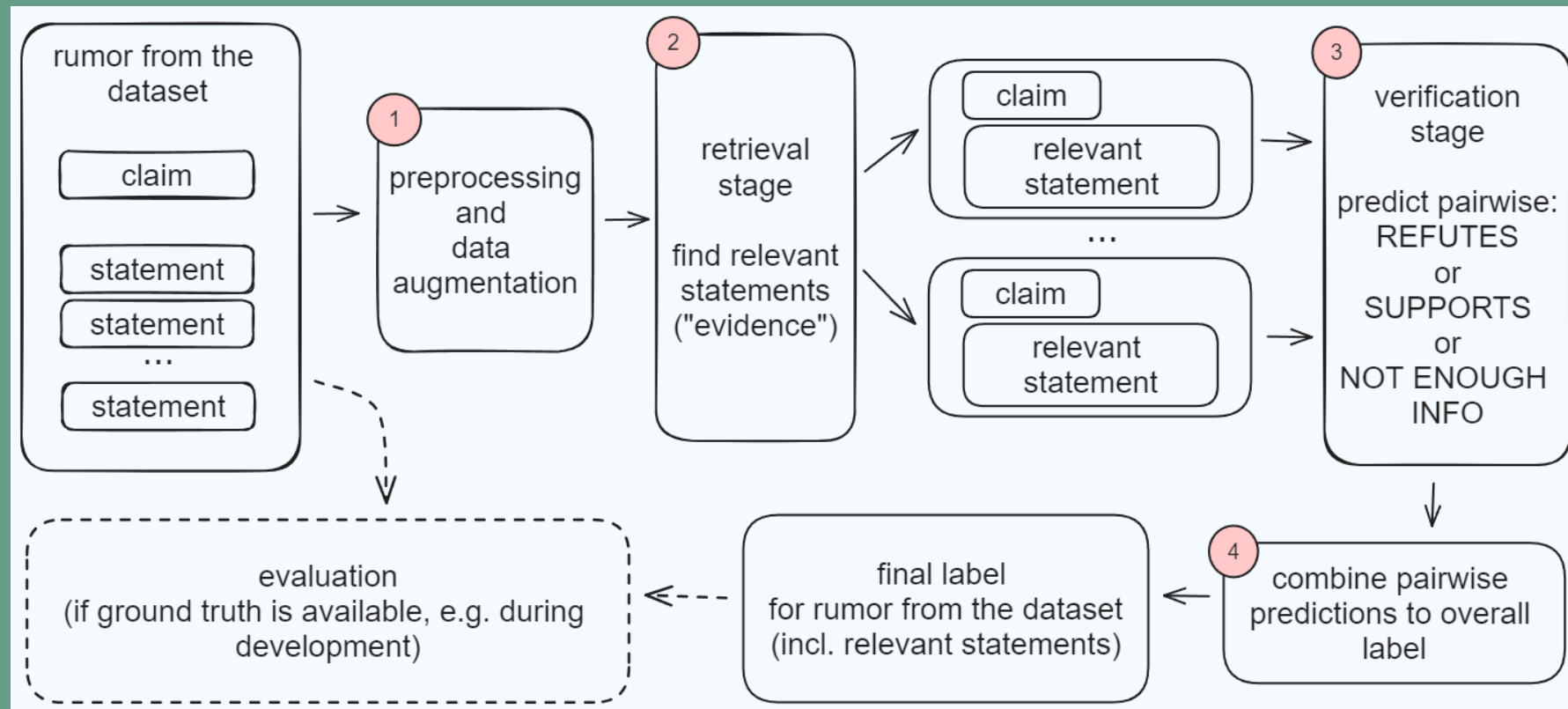
Luis Kolb, Allan Hanbury
@ TU Wien (Austria)



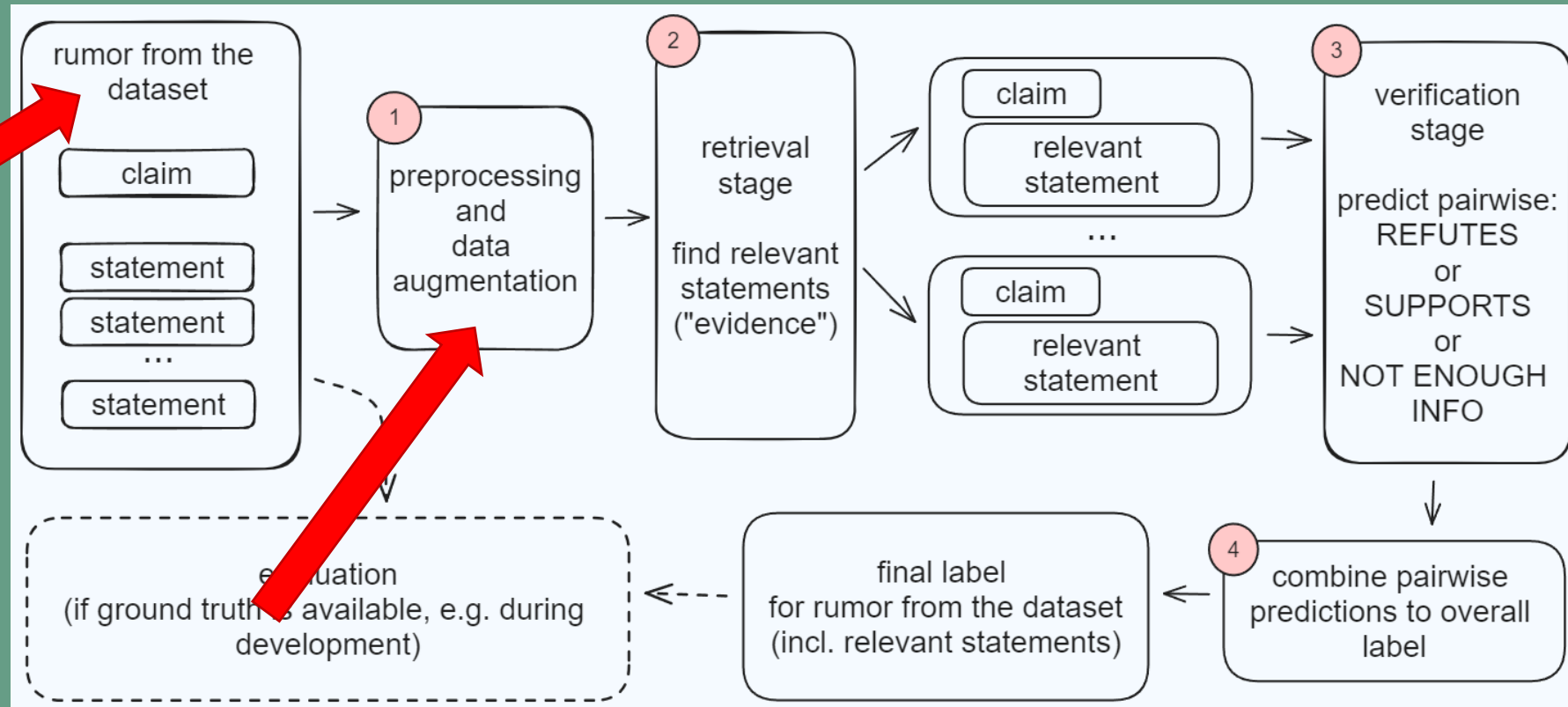
Briefly, an Overview

- We already heard about the task – what was our approach?
- Experiment with Retrieval Approaches
- See if LLMs hold any promise for Verification
- Experiment with NLP Methods for Retrieval & Verification
- We don't train a model, but approach this in a zero-shot fashion
- “Authority” != Government Account (possible, but not necessary)

Our Two-Stage Approach (and task recap)



Our Two-Stage Approach



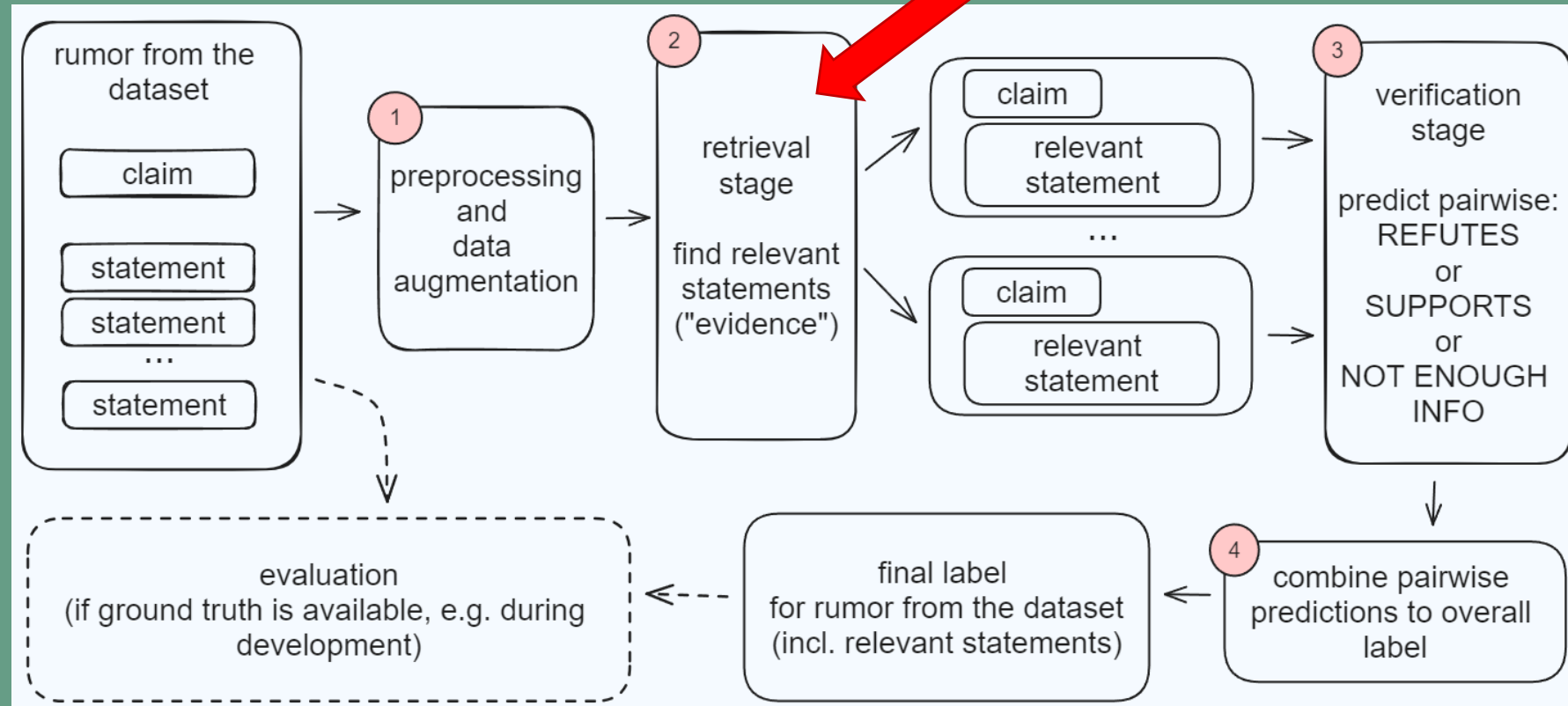
Let's look at an example

```
{
  "id": "AuRED_142",
  "claim": "Naturalization decree in preparation: Lebanese passports for
sale?! https://t.co/UuQ7yMbSWJ https://t.co/Jf1K1NbZJD",
  "timeline": [
    ["https://twitter.com/LBpresidency", "1555424541509386240", "The Information Office of the
Presidency of the Republic: What was published by the French newspaper "Liberation" about the
"selling" of Lebanese passports to non-Lebanese is false and baseless news.'], ...
  ], ...
}
```

Let's look at an example

```
{
  "id": "AuRED_142",
  "claim": "Naturalization decree in preparation: Lebanese passports for
sale?! https://t.co/UuQ7yMbSWJ https://t.co/Jf1K1NbZJD",
  "timeline": [
    ["https://twitter.com/LBpresidency", "1555424541509386240", "The Information Office of the
Presidency of the Republic: What was published by the French newspaper “Liberation” about the
“selling” of Lebanese passports to non-Lebanese is false and baseless news."], ...
  ], ...
}
```

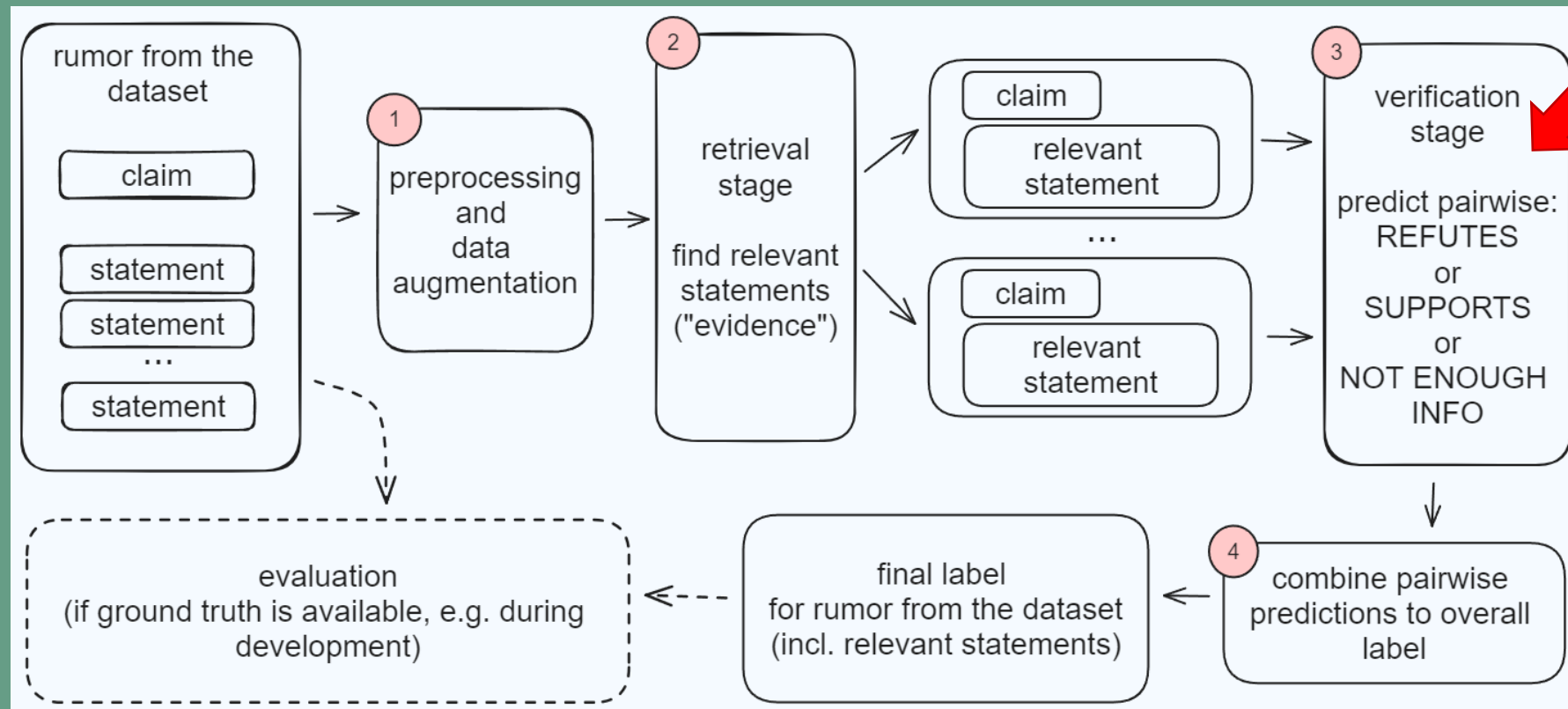
Our Two-Stage Approach



Retrieval Stage – Goals and Approach

- Retrieve relevant tweets from authority account timelines
- Lexical methods: BM25, TF-IDF
- Semantic methods: Embedding Models & Cosine Similarity
- Does preprocessing or additional info about the authority improve Recall?
- Balancing retrieval speed and accuracy → Re-Ranking?

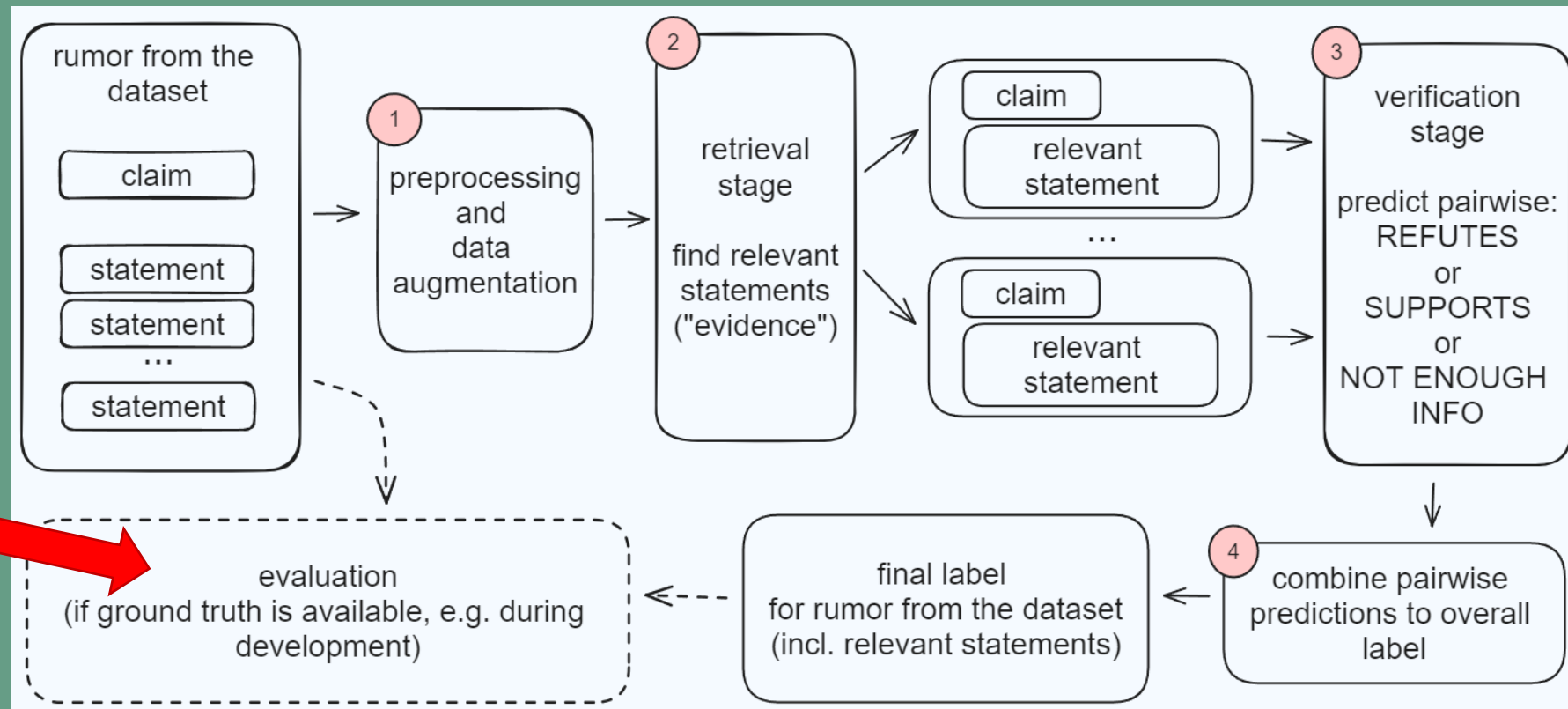
Our Two-Stage Approach



Verification Stage – Strategy and Models

- Verify claim support using retrieved evidence
- Basically Natural Language Inference (NLI)
- Models: BERT Variants fine-tuned for NLI, GPT-4, Llama-3
- Focus on natural language reasoning: how complex would the language model need to be?

Our Two-Stage Approach



Performance Metrics (CheckThat! Task 5)

- Retrieval: Recall@5, Mean Average Precision
- Verification: Macro-F1 (balance of Precision and Recall),
Strict-Macro-F1

Leaderboard – Retrieval (English)

Team	Run Label	MAP	R@5	Retrieval	Preprocess	External Data
IAI Group	secondary1	0.628	0.676			
bigIR	primary	0.604	0.677			
Axolotl	primary	0.566	0.617			
Team DEFAULT	primary	0.559	0.634			
AuthEv-LKolb (ours)	primary	0.549	0.587	Embeddings	False	False
AuthEv-LKolb (ours)	secondary2	0.524	0.563	PyTerrier	True	False
AuthEv-LKolb (ours)	secondary1	0.510	0.619	Embeddings	False	True
	(baseline)	0.335	0.445			

Leaderboard – Verification (English)

Team	Run Label	Macro-F1	Strict Macro-F1	Retrieval	Preprocess Data	External
AuthEv-LKolb (ours)	secondary1	0.895	0.876	Embeddings	False	True
AuthEv-LKolb (ours)	primary	0.879	0.861	Embeddings	False	False
AuthEv-LKolb (ours)	secondary2	0.831	0.831	PyTerrier	True	False
Axlotl	primary	0.687	0.687			
	(baseline)	0.495	0.495			
Team DEFAULT	primary	0.482	0.454			
IAI Group	secondary1	0.459	0.444			
bigIR	primary	0.458	0.428			

Key Findings – Retrieval Stage

- Best* performance via the Terrier implementation of PL2 with some stemming, etc. (divergence-from-randomness lexical method)
 - *at the time of submission
- On average, using Terrier over OpenAI embeddings and cosine similarity yielded a Macro-F1 improvement of +0.0239
- However, the performance gap is small – on the leaderboard, OpenAI embeddings were our best performing retriever method

Key Findings – Verification Stage

- LLMs are indeed capable of the inference step to determine support or refutal based on the claim-evidence pairing
- BERT variants fine-tuned on NLI datasets did not generalize well to this task, but GPT-4-Turbo was much more reliable & performant than Llama3-70b (expected outcome)
- Retrieval only needs to be „good enough“ (for this dataset)

Key Findings – Features

- In our experiments, the best settings were:
 - NO preprocessing done to claim and evidence
 - NO additional author data added to the evidence (display-name & bio)
 - Giving a lot of weight to each piece of evidence was beneficial
 - (ignoring individual “NOT ENOUGH INFO” claim-evidence pairings in the overall rumor labelling decision, as the evidence was likely unrelated to the claim, and we always pass 5 pairings to the verification stage)

Findings since Submission Deadline (1/2)

- A capable enough LLM with correct temperature settings is stable and does not just happen to hallucinate the correct labels
- A re-ranking pipeline of
BM25 >> (%top-50) >> NV-Retriever-V1 >> (%top-5)
performed very well for retrieval (0.71 R@5 on combined train & dev)
- In fact, this pipeline performed the best out of everything we tried later (lexical & semantic methods)

Findings since Submission Deadline (2/2)

- The “weight class” of GPT-4o class of models is a good balance of capability & cost (a bit better than Llama-3.1 variants)

Rank	Macro-F1	Strict-Macro-F1	Retrieval_Method	Verifier_Method
1	0.858523	0.850470	rerank-nv-embed-v1	openai-4o
5	0.810380	0.797802	rerank-nv-embed-v1	openai-4o-mini
9	0.776465	0.768775	rerank-nv-embed-v1	llama3-1-70b
10	0.773538	0.754840	rerank-nv-embed-v1	llama3-1-405b
17	0.652548	0.641708	rerank-nv-embed-v1	llama3-1-8b
21	0.322321	0.322321	rerank-nv-embed-v1	transformers-roberta
23	0.298747	0.280012	rerank-nv-embed-v1	transformers-bart

Verification experiment results on the combined (train & dev) dataset

Caveats & Discussion

- LLMs-as-a-Service are subject to proprietary moderation (and sensitive topics are to be expected when dealing with rumors)
 - Ideally, you would use your own models, and fine-tune them continually on real data
- Scalability would require trade-offs
- This Lab task did not cover finding the authority account timelines our system relies on – this is an entirely separate task
- Multimodality – we did not use images & video content in our approach, though this could be integrated given the multimodal models today

Potential for Future Work

- Multimodality
- Multilingualism
- Integration of more “meta-data” (for example: social graph, or interactions between claim and evidence – was the evidence a response/quote-tweet, or none of these?)
- If we’re already using LLMs – explainability! (via generation of “justifications”)

Conclusion

- We submitted the best performing system to the CheckThat! Lab Task 5 (English)
- LLMs can work as powerful language inference agents
- There is already plenty of distrust in AI (especially LLMs) today!
Science communication is important to build trust in systems like this

Thank you! Questions?



Slides on GitHub 



Retrieve my Contact Info at luiskolb.at